VOL. I · NO. 9 MONDAYS & THURSDAYS

THE LAUNCH REPORT

Rocket launches, reentries and the business of spaceflight, worldwide

MONDAY, NOVEMBER 9, 2026 ALL TIMES EASTERN STANDARD TIME (EST)

THE WEEK IN LAUNCH

Five for five, and Starship lines up a double catch

Five orbital launches flew worldwide since Thursday's issue, and all five reached orbit: a Long March 2C with three Yaogan-51 satellites, Galactic Energy's Ceres-2 on its first successful flight, SpaceX's USSF-432 from Vandenberg, a Soyuz with the Glonass navigation satellite Kosmos 2621, and a Long March 12 with SatNet internet satellites. That brings 2026 to 260 orbital launch attempts and 253 successes, according to GCAT.

Friday, November 13, is shaping up as the busiest day ahead: United Launch Alliance's Vulcan returns with 40 Amazon Leo satellites, and SpaceX says Starship Flight 15 is "targeting to launch as early as Friday, November 13, pending regulatory approval," with an attempt to catch both stages (SpaceX). On the station, Thursday's spacewalk installed the last of eight roll-out solar arrays, and Varda's W-8 capsule came home.

I · RECENT LAUNCHES & RETURNS

Ceres-2 reaches orbit; USSF-432 and Glonass fly

Three Chinese launches, one American and one Russian, all successful, between Thursday morning and Sunday afternoon.

Ceres-2 returns to flight

Galactic Energy's Ceres-2 lifted off at 6:37 p.m. EST Friday (23:37 UTC) from Launch Area 95A at Jiuquan with four commercial Earth-observation and technology-demonstration satellites bound for Sun-synchronous orbit (Launch Library 2). It was the rocket's second launch, "reaching orbit successfully for the first time," Jonathan McDowell noted; the debut in January was one of the year's failures. CelesTrak's catalog shows the four satellites at about 530 by 548 kilometers, inclined 97.6 degrees.

USSF-432: the fourth "R" mission

A Falcon 9 lifted off from Space Launch Complex 4E at Vandenberg at 7:48 p.m. EST Friday (00:48 UTC Saturday) on USSF-432, a classified Space Force mission also identified as R-4, and took a south-easterly trajectory off the California coast, Spaceflight Now reported. The flight was originally scheduled for Nov. 1. Each earlier R launch placed 23 satellites in orbit according to catalog entries; Jonathan McDowell lists the group as part of the Space Force's Space Based Sensing and Targeting program. First stage B1095, on its eighth flight and its first from the West Coast, landed on the drone ship Of Course I Still Love You (SpaceX).

Long March 2C: Yaogan-51

A Long March 2C lifted off at 10:58 a.m. EST Thursday (15:58 UTC) from Launch Complex 3 at Xichang with three Yaogan-51 military remote-sensing satellites (Launch Library 2). CelesTrak lists the trio at about 511 to 524 kilometers, inclined 45 degrees.

Soyuz: Kosmos 2621

A Soyuz-2.1b with a Fregat upper stage lifted off from Site 43/3 at Plesetsk at about 8:22 a.m. EST Saturday (13:22 UTC), according to Launch Library 2. CelesTrak's catalog identifies the payload as Kosmos 2621, a third-generation Glonass-K1 navigation satellite, now in a 19,198 by 19,290 kilometer medium Earth orbit inclined 65.2 degrees.

Long March 12: SatNet Group 29

A Long March 12 lifted off at 3:24 p.m. EST Sunday (20:24 UTC) from Commercial Launch Complex 2 at Wenchang with the 29th group of SatNet internet satellites (Jonathan McDowell). Launch Library 2 lists nine satellites on the flight for the state-owned constellation operated by China Satellite Network Group.

Vulcan, Starship and Electron's 100th crowd Friday night

Every orbital launch with a date between now and November 30. Times are EST with Coordinated Universal Time (UTC) below; NET means no earlier than.

DATE / TIME	VEHICLE	PROVIDER	MISSION / PAYLOAD	SITE	ORBIT	STATUS	SOURCES
Mon Nov 9, 10:03 p.m. 03:03 UTC Nov 10	Falcon 9	SpaceX	USSF-627, a classified U.S. Space Force mission (FAA: "R-5")	SLC-4E, Vandenberg, Calif.	LEO	CONFIRMED	FAA; Launch Library 2
Wed Nov 11, 12:00 a.m. 05:00 UTC	Mir (name provisional)	Agency for Defense Development, South Korea	Demonstration flight of the military small-satellite rocket	Offshore launch platform	LEO	TENTATIVE	Launch Library 2
Wed Nov 11, 6:40 a.m. 11:40 UTC	Long March 2D	CASC	Payload not announced	Launch Area 94, Jiuquan, China	—	TENTATIVE	Launch Library 2
Thu Nov 12, 11:23 p.m. 04:23 UTC Nov 13	Falcon 9	SpaceX	USSF-317, a classified U.S. Space Force mission	SLC-4E, Vandenberg, Calif.	LEO	CONFIRMED	FAA; Launch Library 2
Fri Nov 13, 12:54 p.m. 17:54 UTC	Vulcan VC6L	United Launch Alliance	Vulcan Leo 1 (LV-01; FAA: "KV-01"): 40 Amazon Leo broadband satellites	SLC-41, Cape Canaveral, Fla.	LEO	CONFIRMED	FAA; Launch Library 2
Fri Nov 13, 5:45 p.m. 22:45 UTC	Starship	SpaceX	Flight 15: Booster 22 and Ship 42 with Starlink satellites; first attempt to catch both stages	Orbital Launch Pad 2, Starbase, Texas	LEO	TENTATIVE	SpaceX; Launch Library 2
Sat Nov 14, 12:30 a.m. 05:30 UTC	Electron	Rocket Lab	"The Mountain God Rises": QPS-SAR radar satellite for iQPS; Electron's 100th mission	Launch Complex 1, Mahia, New Zealand	LEO	CONFIRMED	Rocket Lab
Fri Nov 20, 6:00 a.m. 11:00 UTC	Long March 8A	CASC	SatNet low-Earth-orbit communications satellites (group not confirmed)	Commercial LC-1, Wenchang, China	LEO	TENTATIVE	Launch Library 2
NET Sun Nov 22	Ariane 64	Arianespace	VA271 (LE-04): 36 Amazon Leo broadband satellites	ELA-4, Kourou, French Guiana	LEO	PRELIMINARY	Arianespace; Launch Library 2
Tue Nov 24, 12:03 a.m. 05:03 UTC	Soyuz-2.1b	Roscosmos	Progress MS-36 cargo ship to the International Space Station	Site 31/6, Baikonur, Kazakhstan	LEO	TENTATIVE	Launch Library 2
Tue Nov 24, 2:00 a.m. 07:00 UTC	Long March 12	CASC	SatNet low-Earth-orbit communications satellites (group not confirmed)	Commercial LC-2, Wenchang, China	LEO	TENTATIVE	Launch Library 2
NET Tue Nov 24	Long March 2F	CASC	Shenzhou 24 crew to the Tiangong station	Launch Area 91, Jiuquan, China	LEO	PRELIMINARY	Launch Library 2
Thu Nov 26, 3:00 p.m. 20:00 UTC	Spectrum	Isar Aerospace	Third test flight, carrying several satellites including R-Space's AT-Astra	Andøya Spaceport, Norway	SSO	TENTATIVE	Launch Library 2
Sun Nov 29, 10:49 a.m. 15:49 UTC	Falcon 9	SpaceX	Northrop Grumman's Cygnus NG-25 cargo ship to the space station	SLC-40, Cape Canaveral, Fla.	LEO	CONFIRMED	Launch Library 2

CONFIRMED date and time announced by the launch provider, its customer or the FAA. **TENTATIVE** specific date in published schedules or airspace and maritime notices, not yet confirmed by the operator. **PRELIMINARY** target range, or no-earlier-than date or month only.

Abbreviations: SLC, Space Launch Complex; SLC-4E, Space Launch Complex 4 East; ELA-4, Ariane Launch Area 4; LEO, low Earth orbit; SSO, Sun-synchronous orbit; CASC, China Aerospace Science and Technology Corp.; iQPS, Institute for Q-shu Pioneers of Space, a Japanese radar imaging company; USSF, U.S. Space Force.

Date changes

Starship Flight 15. Listed Thursday for November with no specific date; now Friday, November 13, at 5:45 p.m. in Launch Library 2, "pending regulatory approval" (SpaceX). NASASpaceflight reports the target moved one day from November 12, with the FAA ship-catch license still the deciding factor.

Mir demonstration. Tuesday, November 10, at 1:00 a.m.; now Wednesday, November 11, at 12:00 a.m. (Launch Library 2). No reason given.

Electron "The Mountain God Rises." NET Friday, November 13; now NET Saturday, November 14, at 12:30 a.m. Eastern (Rocket Lab). No reason given.

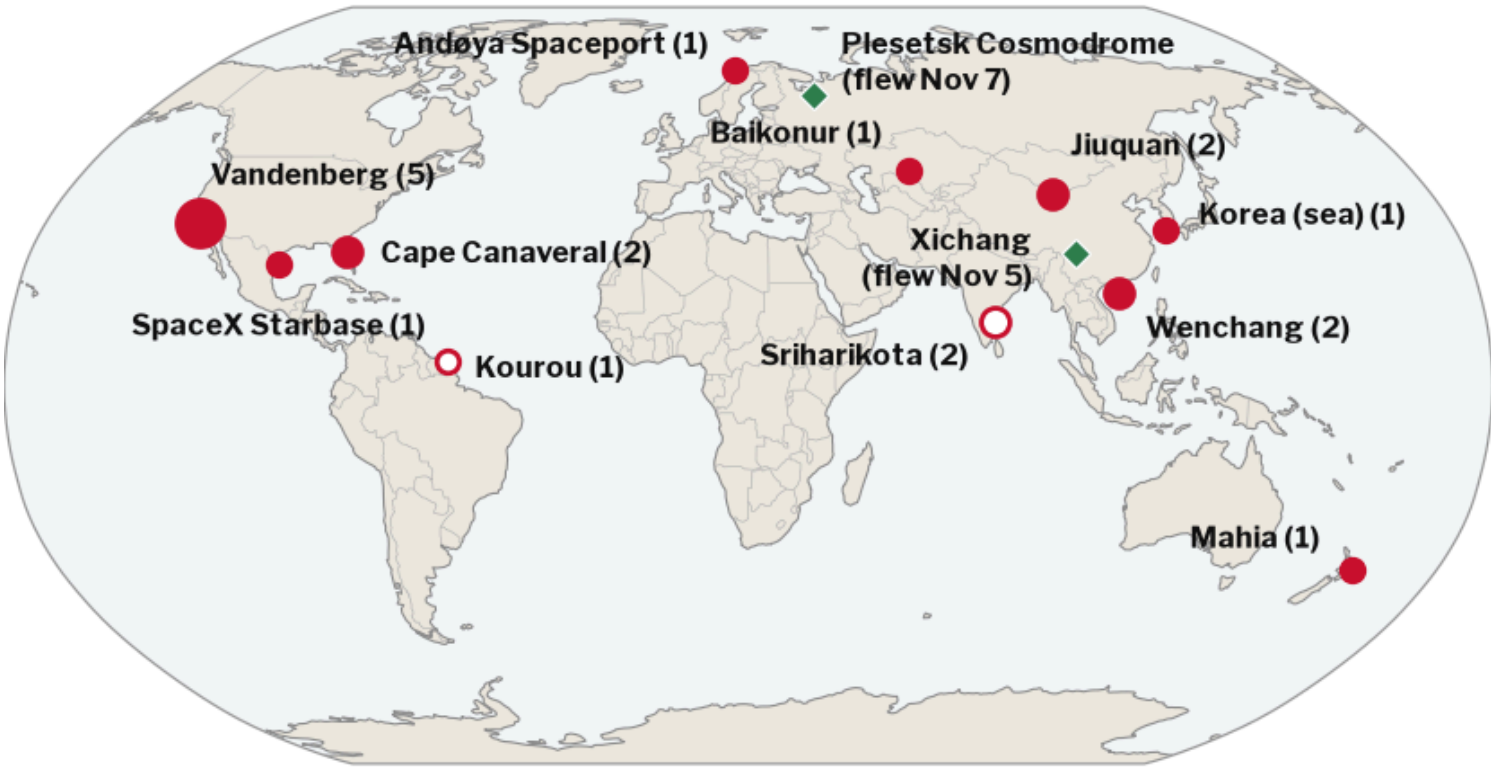
Vulcan Leo 1. Same 12:54 p.m. time, now carried in the FAA plan as "KV-01" with a 12:54 to 2:00 p.m. window (FAA). "One week to launch!" ULA posted Friday.

Also listed for November without a specific date in Launch Library 2: Starlink 15-21, NROL-195 and OneWeb 21 on Falcon 9s from Vandenberg; and ISRO's first uncrewed Gaganyaan test

flight on a Launch Vehicle Mark 3 and a PSLV-XL with the TDS-01 technology demonstration satellite.

QUICK HIT

Amazon Leo livery. Amazon Leo and Arianespace logos are being added to the Ariane 6 fairing that will carry 36 Amazon Leo satellites on VA271 (Arianespace).



Launch sites with orbital launches dated in Launch Library 2 through late November; numbers count launches. Filled: dated launch (Confirmed or Tentative). Hollow: Preliminary. Green diamonds: sites that flew since Thursday with nothing else scheduled this month; Jiuquan, Vandenberg and Wenchang also flew.
MAP: THE LAUNCH REPORT · PAD COORDINATES FROM LAUNCH LIBRARY 2 · NATURAL EARTH

III · TRAJECTORIES & HAZARD AREAS

Airspace windows for two Falcons and Vulcan

The FAA's operations plan lists the times U.S. airspace is set aside; it gives no hazard-area coordinates.

Airspace windows

The FAA Air Traffic Control System Command Center's plan for Monday lists these launch windows, converted from UTC (FAA advisory):

FLIGHT	PRIMARY WINDOW	BACKUP
"R-5" (USSF-627)	**Mon Nov 9, 10:03 p.m.–Tue 2:46 a.m.**	**Tue Nov 10, 9:49 p.m.–Wed 2:32 a.m.**
USSF-317	Thu Nov 12, 11:23 p.m.–Fri 4:06 a.m.	Fri Nov 13, 11:09 p.m.–Sat 3:52 a.m.
"KV-01" (Vulcan Leo 1)	Fri Nov 13, 12:54–2:00 p.m.	Sat Nov 14, 12:32–1:38 p.m.

Starship Flight 15 does not yet appear in the plan (FAA).

Flight 15's planned profile

NASASpaceflight reports a working plan of roughly 24 hours for Ship 42, about 15 to 16 orbits, so its orbit can realign with the reentry corridor back to Starbase and Pad 2. The return involves an overflight of populated areas, which is why SpaceX needs an FAA license modification before the dual catch can fly as planned. Flight 14 lasted just over three hours.

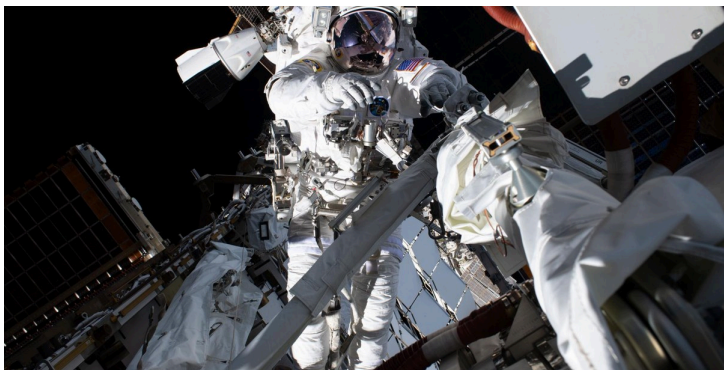
ON THE WIRE

Southeast from Vandenberg. USSF-432 took a south-easterly trajectory off the California coast after Friday's 7:48 p.m. EST liftoff (Spaceflight Now).

Station's power upgrade done; W-8 comes home

The eighth and final roll-out solar array is up, two ESCAPADE probes swing past Earth for Mars, and nine more catalogued objects left orbit.

The last roll-out solar array



NASA astronaut Anil Menon works on the station's Starboard 6 truss during an Aug. 6, 2026, spacewalk to build the mounting kit for the final roll-out solar array.

PHOTO: NASA · PUBLIC DOMAIN

Canadian Space Agency astronaut Joshua Kutryk and NASA astronaut Anil Menon "completed their spacewalk at 3:35 p.m. EST today after installing the eighth and final roll-out solar array," the space station account posted Thursday. The 6-hour, 59-minute outing finished an effort that began in February 2021 and took 18 spacewalks; the eight arrays raise the station's power generation capacity from 160 kilowatts to about 215 kilowatts (NASA).

The array unrolled at about 3:05 p.m. EST, after flight controllers saw lower-than-expected current and held deployment for another orbit so Kutryk could inspect a power connector in darkness, Spaceflight Now reported. Its mounting point on the outermost starboard truss is beyond the reach of Canadarm2, so the pair handed the roughly 750-pound package between them for the last stretch. It was the 286th spacewalk for station assembly and maintenance, Menon's fifth and Kutryk's second. On Friday the crew stowed tools and serviced the suits (NASA).

The next visitors remain Progress MS-36 at 12:03 a.m. November 24 from Baikonur, China's Shenzhou 24 to Tiangong NET November 24, and Cygnus NG-25 at 10:49 a.m. November 29 (Launch Library 2).

ESCAPADE's powered flybys

NASA scheduled the first of its twin ESCAPADE spacecraft to swing past Earth with a roughly 9-minute "trans-Mars injection burn" around 5 a.m. EST Saturday, and the second around 6 a.m. EST Monday, each adding more than 1,000 miles per hour; arrival at Mars is set for Sept. 7 and Sept. 9, 2027 (NASA). The agency says no Mars-bound spacecraft has completed a powered flyby. Rocket Lab, which built the probes Blue and Gold, wrote Thursday night that they "are about to execute a trans-Mars Injection maneuver to propel themselves to Mars."

Confirmed reentries

CelesTrak's satellite catalog lists these objects as decayed since Thursday's issue. Dates are UTC.

OBJECT (CATALOG NO.)	LAUNCHED	DECAYED	LAST ORBIT, KM
W-Series 8 (100883)	**Oct 1, 2026**	**Nov 5**	**311 × 906**
Starlink-2298 (48022)	Mar 24, 2021	Nov 5	152 × 159
Starlink-1420 (45695)	Jun 4, 2020	Nov 6	152 × 154
Unnamed U.S. payload (60203)	Jul 4, 2024	Nov 6	174 × 179
Starlink-36253 (68028)	Mar 1, 2026	Nov 6	175 × 192
Starlink-1338 (45532)	Apr 22, 2020	Nov 7	139 × 152
Starlink-1730 (46563)	Oct 6, 2020	Nov 7	157 × 160
MENUT (55010)	Jan 3, 2023	Nov 7	198 × 209
Starlink-4607 (53740)	Sep 5, 2022	Nov 8	199 × 202

The catalog has also added Swift's Nov. 4 decay date (28485), along with two more Nov. 4 entries: Starlink-32664 and an unnamed Chinese payload from 2022 (CelesTrak). W-Series 8 is Varda Space Industries' W-8 capsule, which fired its final deorbit burn at 8:10 a.m. EST Thursday carrying hypersonic data for government customer payloads (Varda). "W-8 is back on Earth. Let's do it again in a few weeks," the company posted that evening.

ON THE WIRE

Mars in September 2027. After their powered flybys, the two ESCAPADE spacecraft are due at Mars on Sept. 7 and Sept. 9, 2027 (NASA).

Starship readies a double catch; New Glenn gets a third stage

SpaceX waits on an FAA license change, Blue Origin and AST SpaceMobile unveil new hardware, NASA widens its Moon Base contracts, and the FAA proposes more Part 450 changes.

SpaceX: Flight 15

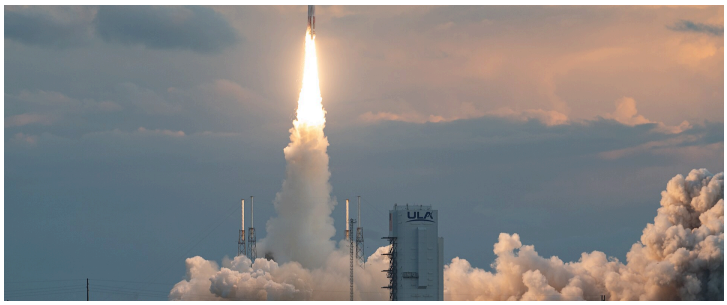
"SpaceX will attempt to catch both the Super Heavy booster and Starship on the upcoming flight. If both are successful, it will mark the first time that all parts of an orbital rocket return to Earth intact," Elon Musk (@elonmusk) wrote at 10:47 a.m. EST Friday. No ship has yet been caught, and Flight 15 would also bring the first Block 3 booster catch and the first catch at Pad 2, NASASpaceflight reported. Booster 22 and Ship 42 have completed full-duration static fires; the ship still needs its flight termination charges and a batch of Starlink satellites. The Flight 14 license order covered that flight alone, so SpaceX needs FAA approval of a license modification, including permission to return Ship 42 to the tower.

The Flight 16 pair is lined up for the first Starship launch from Launch Complex 39A at Kennedy Space Center, a debut that now looks more likely early in 2027 (NASASpaceflight).

Blue Origin: GS3 and Northstar

Blue Origin chief executive Dave Limp "unveiled GS3, a third stage that significantly expands New Glenn's ability to deliver high-energy and deep-space missions, and Northstar, a new booster engine," the company posted Thursday. Powered by a BE-7 engine, GS3 is quoted at up to 8 metric tons to geostationary orbit and 10 to trans-lunar injection on today's 7×2 New Glenn, and 14 and 20 on the larger 9×4; it is under contract for a 2028 cislunar mission for an unnamed government customer. Northstar burns liquid oxygen and liquefied natural gas at 715,000 pounds of thrust (NASASpaceflight).

United Launch Alliance: Vulcan's return



Vulcan lifts off from Space Launch Complex 41 at Cape Canaveral Space Force Station on its second certification flight on Oct. 4, 2024.

U.S. SPACE FORCE PHOTO BY AIRMAN 1ST CLASS COLLIN WESSON · PUBLIC DOMAIN

"One week to launch!" ULA posted Friday about Vulcan Leo 1, which carries 40 Amazon Leo satellites on a VC6L with six solid rocket boosters. The FAA plan now holds a 12:54 to 2:00 p.m. EST window on Friday (FAA). It is Vulcan's first flight since nozzle failures on the Northrop Grumman 63XL motors grounded the rocket after the February USSF-87 mission; ULA says the boosters now use an improved nozzle design (Spaceflight Now).

AST SpaceMobile's Blacklift

AST SpaceMobile and Firefly Aerospace announced a $1.3 billion agreement for Firefly engines to power Blacklift, a 20-ton-class rocket AST plans to fly from 2028 to launch five of its BlueBird satellites at a time, Spaceflight Now reported. Firefly will deliver 27 Miranda and nine vacuum Vira engines for the first three rockets and license AST to build more. AST plans an unveiling event in New York on Thursday, November 12.

Pads, budgets and a slip

Stoke Space. Space Launch Delta 45 granted Stoke right of entry to land north of its Space Launch Complex 14, at the old LC-15 site, to study a second Nova pad; Nova Pathfinder is targeted for around mid-2027 (NASASpaceflight).

Canada. The federal budget released Thursday funds the Launch the North initiative and plans to move Canadian rockets up to medium lift, with smaller rockets expected to fly from east-coast spaceports in 2028 (Payload).

Rocket Factory Augsburg. RFA now targets 2027 for the first RFA ONE launch from SaxaVord in Scotland, after the vehicle was destacked in July for an investigation (European Spaceflight).

QUICK HIT

Neutron dome. Rocket Lab says Neutron's first-stage forward dome has come off the automated fiber placement machine and been cured; analysis and material tests come next (Rocket Lab).

NASA: Moon Base awards

NASA picked eight companies, Blue Origin, Firefly Aerospace, Intuitive Machines, ispace-U.S., Lockheed Martin Space, Northrop Grumman Space Systems, SpaceX and Voyager Technologies, for CLPS 2, the next phase of Commercial Lunar Payload Services, a 10-year contract with a combined maximum value of $10 billion. It also raised the current CLPS contract from $2.6 billion to $4.2 billion and moved ahead with PROMISE, a nuclear-powered South Pole rover (NASA).

FAA: three more Part 450 proposals

Monday's Federal Register carries three FAA proposed rules to ease the commercial launch and reentry licensing rule, Part 450, each open for comment until December 9: a conditional expected-casualty exception for failures ending in an uncontrolled atmospheric return (docket FAA-2026-9942), clearer flight safety limits objectives (FAA-2026-9937), and when an operator need not abort a flight at the end of flight safety limits (FAA-2026-9943). A final rule of technical corrections takes effect the same day (FAA-2026-7035).

Still open. Comments on the Federal Communications Commission's space-licensing overhaul (SB Docket No. 25-306) are due November 25 (Federal Register), and Starbase scoping comments on up to 456 Starship launches a year are due November 30, with meetings November 17 to 19 (Federal Register).

VI · BY THE NUMBERS

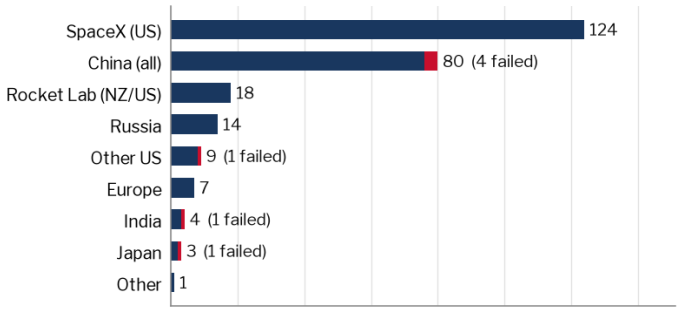
260 orbital launch attempts so far in 2026

Five successes since Thursday lifted the year-to-date count, with GCAT now current through Sunday's Long March 12.

260	253	124	80	14
orbital launch attempts worldwide through Nov 8	reached orbit	by SpaceX, all successful	by China, four failures	by Russia, all successful

The count

Jonathan McDowell's General Catalog of Artificial Space Objects (GCAT) records 260 orbital launch attempts in 2026 through Sunday, 253 of them successful. SpaceX leads with 124, all successful, followed by China with 80, Rocket Lab with 18 and Russia with 14; other U.S. providers have flown nine, Europe seven, India four, Japan three and South Korea one.



Orbital launch attempts in 2026 by provider or country. Navy: success. Red: failure or partial failure. "Other" is South Korea.

CHART: THE LAUNCH REPORT · GCAT THROUGH NOV 8

The failures

According to GCAT, seven launches have failed this year: India's PSLV on January 12; China's Long March 3B on January 16, Ceres-2 on January 17, Tianlong-3 on April 3 and Long March 7A on August 10; Japan's Kairos on March 5; and Blue Origin's New Glenn on April 19, recorded as a partial failure. Ceres-2, the third of those failures, reached orbit on its return Friday (Jonathan McDowell).

The week ahead

Seven orbital launches have specific dates between Monday and Saturday morning (Launch Library 2; FAA): USSF-627 tonight, South Korea's Mir demonstration and a Long March 2D on Wednesday, USSF-317 late Thursday, Vulcan Leo 1 and Starship Flight 15 on Friday, and Electron's 100th mission early Saturday.

Takeaways

1. All five launches since Thursday succeeded, including Ceres-2's first trip to orbit.
2. 2026 stands at 260 orbital attempts and 253 successes.
3. Starship Flight 15 targets Friday with a first try at catching both stages, pending an FAA license change.
4. The station's eighth and final roll-out solar array is installed, lifting capacity to about 215 kilowatts.
5. FAA comments on three new Part 450 proposals are due December 9.

KEY DATES AHEAD (EST)

Mon Nov 9	USSF-627, 10:03 p.m.
Wed Nov 11	Mir demonstration, 12:00 a.m.; Long March 2D, 6:40 a.m.
Thu Nov 12	USSF-317, 11:23 p.m.
Fri Nov 13	Vulcan Leo 1, 12:54 p.m.; Starship Flight 15, 5:45 p.m.
Sat Nov 14	Electron's 100th mission, 12:30 a.m.
Tue Nov 24	Progress MS-36, 12:03 a.m.
Wed Dec 9	Part 450 comments due

SOURCES

Schedules: Launch Library 2 (The Space Devs), NASASpaceflight, FAA operations plan advisory. Agencies: NASA, FAA, FCC, Federal Register. Reporting: Spaceflight Now, NASASpaceflight, Payload, European Spaceflight. Tracking: CelesTrak, GCAT (Jonathan McDowell). X posts: @SpaceX, @elonmusk, @RocketLab, @ulalaunch, @blueorigin, @Arianespace, @Space_Station, @VardaSpace, @planet4589. Photos: NASA and U.S. Space Force (public domain). Maps: Natural Earth (public domain).

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